PARAM DESAI

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Work Experience

Applied AI Engineering Intern, DYU Labs Pvt. Ltd., Ahmedabad, India *May – Jul 2026*

- Built production AI systems for a renewable-energy intelligence platform serving EPC companies, O&M teams, and government solar projects, with deployments across customers including ISRO, NTPC, and Aditya Birla Group.
- Designed **aALoK**, a LangGraph-powered conversational assistant using RAG, Whisper, pgvector, and NL-to-SQL workflows, achieving 2.5-5 s end-to-end query latency.
- Developed a per-inverter LightGBM forecasting pipeline using Copernicus ERA5/ECMWF weather data, replacing plant-level forecasting with device-specific prediction models.
- Implemented automated drift monitoring, feedback-driven retraining, and AWS deployment, achieving **3.15% Clean MAPE** across 75+ active inverters.

Publications

Desai, P. *et al.* (2026). **FIDES: A Federated Intelligence and Detection with Quantum Security for Financial Institutions.** *2026 International Conference on Computer, Information and Telecommunication Systems (CITS)*, 1–6.

Shah, P., Desai, P. *et al.* (2025). **Q-PhishNet: Quantum-Secured Explainable Machine Unlearning for Phishing Detection in IoT Networks.** *2025 IEEE Globecom Workshops (GC Wkshps)*, 2265–2270.

Desai, P. *et al.* (2026). **Explainable AI and Quantum Security for Smart Homes Network Attack Classification. In: Bansal, J.C., Borah, S., Hussain, S., Salhi, S. (eds) Computing and Machine Learning.** CML 2025. Lecture Notes in Networks and Systems, vol 1614. Springer, Singapore.

Desai, P. *et al.* (2026). **Q-TBNet: Machine Unlearning-based Privacy-First Medical Imaging Framework for TB Detection.** *Procedia Computer Science*, *290*, 625–632.

Mehta, D., Soni, V., Desai, P. *et al.* (2025). **Quantum-Assisted XAI-Driven DL Framework for FDI Detection in Connected Autonomous Electric Vehicles Underlying 6G.** *2025 5th Intelligent Cybersecurity Conference (ICSC)*, 349–354.

Koradiya, N., Shah, A., Benslimane, A., Mahida, N., Desai, P., Gupta, R., & Tanwar, S. (2025). **Q-ShielD: Quantum-Enhanced Secure Framework for Autonomous Vehicles Communication.** *2025 IEEE 101st Vehicular Technology Conference (VTC2025-Spring)*, 1–6.

Upadhyay, R., Desai, P. *et al.* (2025). **Quantum-based Edge Intelligence Framework for Wearable Health IoT Device Networks.** *2025 IEEE International Conference on E-Health Networking, Application & Services (Healthcom)*, 7–12.

Submitted Manuscript

Desai, P. *et al.* (2026). **BHOOMI: A Grid-Based Framework for Urban Environmental Forecasting & Spatial Intervention Planning.** Submitted to the ASME *Journal of Computing and Information Science in Engineering* (JCISE)

Research Experience

Undergraduate Research Assistant, ST Lab – Dr. Sudeep Tanwar's Research Group, Ahmedabad, India

Advisors: Dr. Sudeep Tanwar and Dr. Rajesh Gupta *Oct 2024 - Dec 2025*

- Conducted research across Explainable AI, Machine Unlearning, Federated Learning, AI Security, Quantum Key Distribution, and Healthcare AI, resulting in **7 published papers** across **5 IEEE**, **1 Elsevier**, and **1 Springer** venue, including **3 first-author papers**.
- Identified research gaps, designed system architectures, implemented experimental pipelines, and authored manuscripts spanning trustworthy and privacy-preserving intelligent systems.
- Presented research at IEEE ICSC 2025 (Tampa, USA) on a quantum-assisted explainable framework for false-data-injection detection in connected autonomous vehicles.

Achievements

- **Certificate of Scholar:** Awarded in Semesters I and II for showcasing academic excellence.
- **International Conference Presenter:** Presented my research at IEEE ICSC 2025 in Florida, USA, on quantum-assisted explainable AI for secure connected autonomous vehicle systems.
- **IIT Madras:** Completed requirements for Foundation Level & Diploma in Programming.
- **Runner-Up, AETRIX 2026:** National-level hackathon hosted by Pandit Deendayal Energy University.

Technical Projects

Xeroww - Intelligent QuantFinance | *Kafka, Spark, Apache Iceberg, FastAPI* *Sept 2026*

- Architected a distributed, event-driven platform for Indian equities using CQRS and domain events, integrating real-time market analysis, deterministic technical signals, and point-in-time historical replay without look-ahead bias.
- Engineered a streaming analytics pipeline with **Kafka, Spark Structured Streaming, Apache Iceberg, PostgreSQL, Redis, and FastAPI**, supporting low-latency event processing, explainable fuzzy inference, and reproducible quantitative analysis.

FC Analytics – Football WC'26 Simulation and Analytics | *React, FastAPI, Websockets* *Jun 2026*

- Developed a probabilistic football analytics engine combining **Poisson distributions, Elo ratings, and dynamic power indexes**, with interpretable attacking, defensive, form, and team-strength indicators.
- Implemented a **Monte Carlo simulation engine** supporting up to **10,000 runs** of the 48-team World Cup, alongside a real-time fantasy auction system using WebSockets.

E-commerce Cloud Deployment Automation | *Terraform, AWS EKS, Amazon ECR, Docker* *Oct 2026*

- Co-developed a CI/CD pipeline that provisions AWS infrastructure with **Terraform**, publishes container images to Amazon **ECR**, and deploys a frontend and two backend services to **EKS**. Fixed frontend dependency installation and corrected service load balancer configuration.
- Added a GitHub Actions teardown workflow that cleans up Kubernetes resources and AWS load balancers before running terraform destroy.

Education

Institute of Technology, Nirma University, Ahmedabad, India *2023 – 2027*

B.Tech. in Computer Science and Engineering; Minor in Applied Mathematics

CGPA: 8.33/10

Community Service

Volunteer Teacher, Vidhyadaan Foundation, Ahmedabad, India *Jun – July 2024*

- Taught Mathematics, English, and Science to 50 underprivileged children from nearby worker communities through structured classroom sessions.

Clubs & Organizations

Treasurer, Computer Society of India - Nirma University, Ahmedabad, India *Aug 2025-Sept 2026*

- Hosted HACKaMINeD 2026, a national 48-hour hybrid hackathon, in collaboration with SUNY Binghamton University. It had 2,000+ participants from 150+ universities across 20+ states, supported by 13 sponsors and a prize pool exceeding Rs. 500,000.
- Managed finance, logistics, and organisation of HackNUThon 6.0, a 48-hour national-level offline hackathon attracting 1,200+ participants from across India.

Technical Skills

Programming Languages: Python, C++, C, TypeScript, JavaScript, SQL
AI/ML: PyTorch, LangGraph, NLP, Machine Unlearning, Explainable AI (XAI)
Frameworks: FastAPI, Flask, React, Apache Iceberg, Apache Spark, Apache Kafka, Prometheus, Grafana, Docker
Competitive Programming: 300+ Problems Solved across platforms
Languages: English, French (Academic), Hindi, Gujarati

Links

GitHub: https://github.com/paramdesai24
Portfolio: https://paramdesai24.me
Publications: https://scholar.google.com/citations?user=qj1nZ4EAAAAJ